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ISTA 350
Final Project Images

Dataset: <https://pokedexdb.net/pokedex/all>

Filename used to generate images: pokemon_final_draft.py

The dataset chosen is a collection of every Pokémon from generations 1-9. It includes information such as their Pokédex number, name, type, total stats, and individual stat values like HP, Attack, Defense, Special Attack, Special Defense, and Speed.

The columns are labeled as:

#, Name, Type, Total, HP, Attack, Defense, Sp. Atk, Sp. Def, Speed

This data is useful for analyzing the value of Pokémon types and stats, as many competitive players look for optimal builds when competing in tournaments. Some builds prioritize stats like attack, speed, or defense, depending on the strategy being used and on any restrictions that the tournament may have in place. Using this dataset, the competitors can easily identify which Pokémon best match certain stat distributions and desired play styles.

I chose this dataset because I have always been a Pokémon fan and have played about 95% of the games that have been released. Although I do not play it in a competitive manner, it is still interesting to me to see what types are typically stronger than others, who the strongest pokémon are, and how different stats affect a Pokémon's survivability or overall strength.

IMPORTANT CONTEXT:

- Pokémon with a total stat of 180-400 are considered low tier (usually first stage Pokémon)
- Pokémon with a total stat of 400-500 are considered mid tier (usually middle stage Pokémon, or low level final stage Pokémon)
- Pokémon with a total stat of 500-600 are considered high tier (fully evolved Pokémon, starters)
- Pokémon with a total stat of 600+ are usually mega evolved Pokémon, pseudo-legendary Pokémon, or legendary Pokémon (an example of a legendary Pokémon is Palkia: the ruler of space)

Image 1 - Glass Cannon Test

Function from pokemon_final_draft.py that created Image 1: def glass_cannon(df)

In Pokémon, there is a term commonly used to describe Pokémon with high Attack + Speed stats, but also have low HP + Defense + Sp. Defense stats. These Pokémon are called “glass cannons”, as they have very high offensive power but are easily knocked out by opponent Pokémon.

To determine if there is an actual correlation between high offensive power + low defensive capabilities, I combined Attack + Speed stats to be the offensive variable and combined HP + Defense + Sp. Defense to be the defensive variable. I used a scatter plot with a fitted linear regression line to visualize the correlation between the two variables, which showed that there was a positive correlation. This indicates that defense stats aren't typically negatively affected by offensive stats, and that glass cannons are the outliers whose defense was negatively affected.

I found this very interesting because I tend to prefer Pokémon that have high offensive capabilities. One of my personal favorites is Greninja, and sadly they are one of the outliers and are labeled as a glass cannon. I had already known that Greninja was a glass cannon based on my experiences in a battle where Greninja would get knocked out in 1-2 hits by my opponent, so I had subconsciously established a bias that all Pokémon similar to Greninja would also be considered glass cannons. Seeing that Greninja is an outlier to the correlation, I can determine that my personal bias was incorrect and I had simply gotten unlucky by choosing one of the Pokémon who actually fit in the glass cannon description.

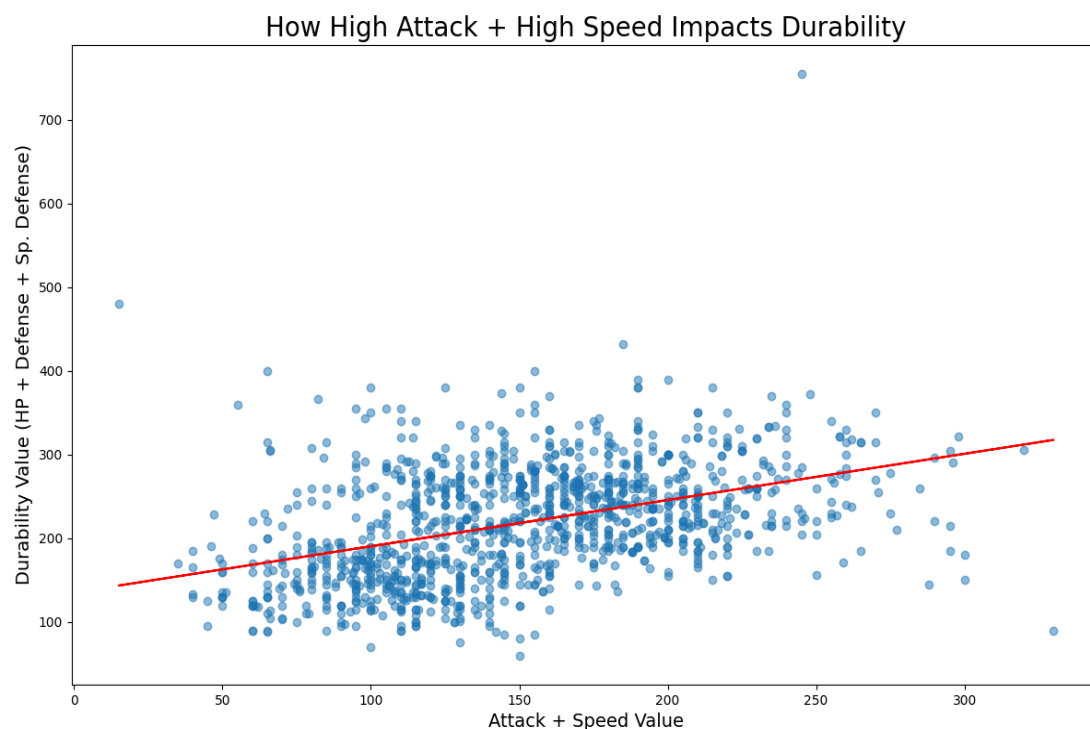


Image 2 - Best Primary Type

Function from `pokemon_final_draft.py` that created Image 2: `def primary_stats(df)`

In Pokémon, there are numerous types and many Pokémon also have two types. Pokémon types essentially work as a very complicated version of rock paper scissors, where fire beats grass, water beats fire, and grass beats water. This raises the question, since all Pokémon have a total stat that can be compared; which primary Pokémon type is the strongest?

To solve this, I first began by separating the primary type from the Type column. The primary type is the first type that appears. So Venusaur is a Grass/Poison type Pokémon, and its primary type would be Grass. Once I separated the primary type for all of the Pokémon, I grouped by primary type and calculated the mean of the total stat.

Once I had my primary types separated and their total stats calculated, I created a bar graph that would allow us to visualize the difference between each Pokémon type. The bar graph showed that dragon types had the highest average total stats, having a value of 531. Surprisingly, steel types came in second with a value of around 490. This shocked me as I had thought a starter type (grass, water, fire) would have been second, as the starter Pokémon tend to be stronger than other Pokémon you encounter throughout the games. This leads me to believe that this could have occurred due to games where the primary types of fire, water, or grass were not as present as the other types (for example in Sinnoh, there's only a few fire types available throughout the whole region).

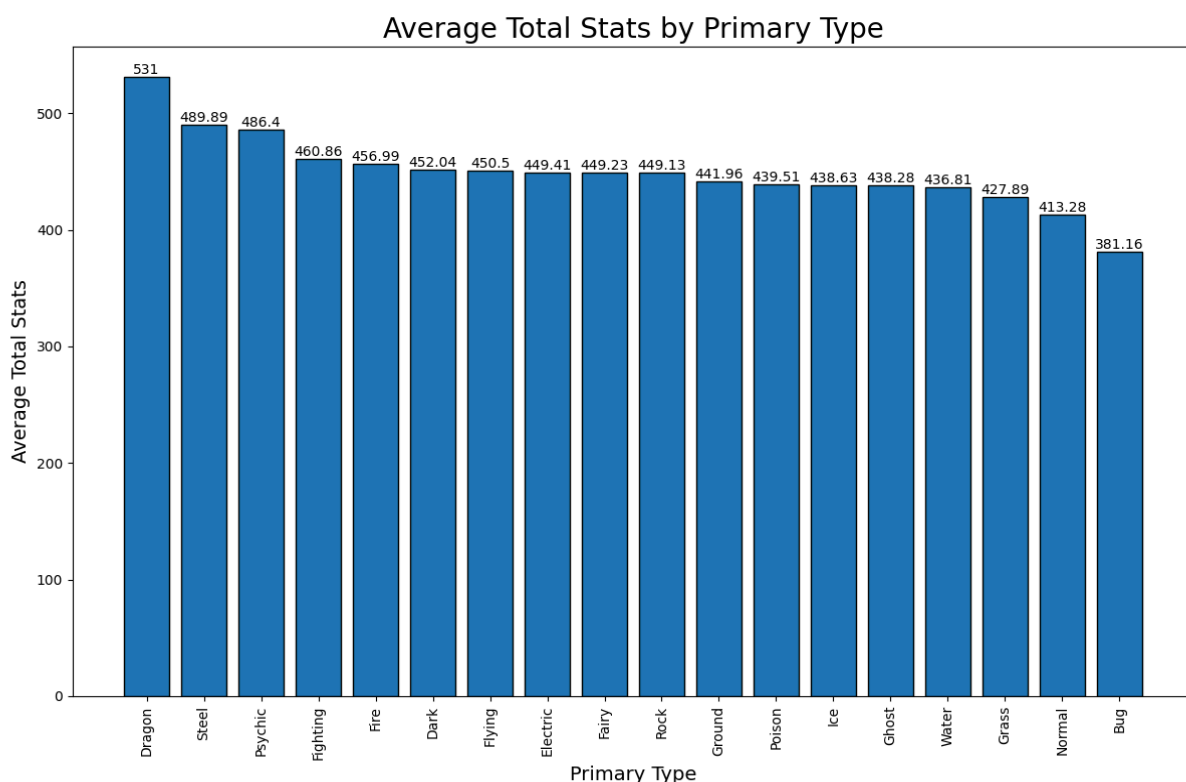


Image 3 - Top 10 Strongest Pokémon

Function from `pokemon_final_draft.py` that created Image 3: `def top_10_pokemon(df)`

There are a total of 1,025 Pokémon throughout generations 1-9. While on my quest to catch 'em all and beat the Pokémon league, I did occasionally wonder if it is actually worth catching all 1,025. Because each game limits the Pokémon you encounter, and your team is limited to a size of 6, there is sadly no room for Pokémon who do not pull their own weight. I might be sounding a little too much like Team Rocket (the bad guys who steal Pokémon), but I do want to find the strongest Pokémon to make sure becoming the Pokémon champion is as easy as possible.

To achieve this, I created a `top10` dataframe that was a sorted version of the `pokemon` dataframe. It was sorted by total stat and was limited to a size of 10 in order to obtain the top 10 Pokémon. Once I had the Pokémon, I created a horizontal bar graph to display the results of who the top 10 were.

Unsurprisingly, each of the top 10 Pokémon were all legendary Pokémon. However, what surprised me was the drastic difference between 1st and 2nd place. The strongest Pokémon according to total stats was Eternatus Eternamax, with a value of 1,125. The second strongest was a tie between Mega Mewtwo Y, Mega Rayquaza, and Mega Mewtwo X who all had a value of 780. A difference of 345 was mind blowing, especially because I had never before seen a Pokémon have a total stat value exceed 1,000 before. This highlights the evolution of power systems within Pokémon, as mega evolution was introduced in generation 6 (2013-2014) and Eternatus Eternamax was not introduced until generation 8 (2019).

